MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **Periodic Test Paper 13 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. If the common difference of an AP is 3, then $a_{20} - a_{15}$ is: [1]

(A) 5 (B) 15 (C) 8 (D) 12

Q2. The nth term of the AP $\frac{1}{m}, \frac{1+m}{m}, \frac{1+2m}{m}, \ldots$ is: [1]

(A) $\frac{1+nm}{m}$ (B) $\frac{1+(n-1)m}{m}$ (C) $\frac{n+m}{m}$ (D) $\frac{m+(n-1)}{m}$

Q3. How many two-digit numbers are divisible by 3? [1]

(A) 25 (B) 30 (C) 32 (D) 33

Q4. What is the common difference of an AP in which $a_{21} - a_7 = 84$? [1]

(A) 4 (B) 5 (C) 6 (D) 7

Q5. **Assertion (A):** The sum of first n terms of an AP with first term a and last term l is $S_n = \frac{n}{2}(a + l)$.
Reason (R): Substituting $l = a + (n-1)d$ into $\frac{n}{2}(a + l)$ yields $S_n = \frac{n}{2}[2a + (n-1)d]$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. In an AP, given $a = 2, d = 8, S_n = 90$, find the number of terms n and the nth term a_n. [2]

Q7. Find the 10th term from the end of the AP $8, 10, 12, \ldots, 126$. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. If the 9th term of an AP is zero, prove that its 29th term is double its 19th term. [3]

Q9. The 19th term of an AP is equal to three times its 6th term. If its 9th term is 19, find the AP. [3]

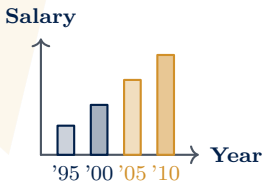
SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. A spiral is made up of successive semicircles with centres alternately at A and B, starting with centre at A, of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, ... What is the total length of such a spiral made up of 13 consecutive semicircles? Take $\pi = \frac{22}{7}$. [5]

Q11. If the sum of the first n terms of an AP is given by $S_n = 3n^2 + 5n$ and its mth term is 164, find the value of m. Also, find the 20th term of this AP. [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Career Increments & Salary Amortization (Subba Rao's Job)



Subba Rao joined a government organization in the year 1995 at an initial starting annual salary of Rs. 5,000. Under the civil service compensation rules, he received a fixed uniform annual increment of Rs. 200 each year. His career earnings and progressive annual compensations form an arithmetic progression.

Based on the above employment record, answer the following questions:

(i) Write the arithmetic progression representing Subba Rao's annual salary, stating the first term a and common difference d. [1]

(ii) What was his annual salary in the year 2005 (after 10 completed years)? [1]

(iii) In which year did his annual salary reach Rs. 7,000? [3]

OR

(iv) Calculate the total cumulative earnings received by Subba Rao during his first 15 years of service. [3]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★